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Electronic smoking article

Abstract

An electronic smoking article includes a liquid supply region including liquid material and a heater-wick element operable to wick liquid material and heat the liquid material to a temperature sufficient to vaporize the liquid material and form an aerosol. The heater-wick element comprises a plurality of fused metal beads or particles.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 17/212,334, filed Mar. 25, 2021, which is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 16/436,158, filed Jun. 10, 2019, which is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 15/997,324, filed Jun. 4, 2018, which is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 14/185,299, filed Feb. 20, 2014, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 61/768,080, filed on Feb. 22, 2013, the entire contents of each of which are incorporated herein by reference.

SUMMARY

(1) An electronic smoking article is provided which includes a heater-wick element which wicks liquid and heats the liquid material to produce an aerosol or “vapor.” The heater-wick element preferably comprises a plurality of metal beads or particles fused together into a frit of a desired shape. The heater-wick element includes a wicking portion and a heatable portion, which are integrally formed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a top planar view of an electronic smoking article according to a first embodiment;
- (2) FIG. 2 is a side cross-sectional view of the electronic smoking article shown in FIG. 1 including a heater-wick element as described herein;
- (3) FIG. 3 is an enlarged view of the heater-wick element of FIG. 2;
- (4) FIG. 4 is an enlarged view of an electrical connection for a heater-wick element as described herein; and
- (5) FIG. 5 is an enlarged view of another embodiment of a heater-wick element.

DETAILED DESCRIPTION

(6) Referring to FIGS. 1 and 2, an electronic smoking article (article) **60** is provided and comprises a replaceable cartridge (or first section) **70** and a reusable fixture (or second section) **72**, which in the preferred embodiment are coupled together at a threaded connection **205** or by other convenience such as a snug-fit, detent, clamp and/or clasp. Generally, the second section **72** includes a puff sensor **16** responsive to air drawn into the second section **72** via an air inlet port **45** adjacent the free end or tip of the electronic smoking article **60**, a battery **1** and control circuitry. The disposable first section **70** includes a liquid supply region of **22** including liquid and a heater-

wick element **14** that wicks liquid from the liquid supply region **22** and heats the liquid to form an aerosol in a central air channel **21**. Upon completing the threaded connection **205**, the battery **1** is electrically connected with the heater-wick element **14** of the first section **70** upon actuation of the puff sensor. Air is drawn primarily into the first section **70** through one or more air inlets **44**.

(7) In the preferred embodiment, once the liquid of the cartridge is spent, only the first section **70** is replaced. An alternate arrangement includes a layout where the entire electronic smoking article **60** is disposed once the liquid supply region is depleted. In such case the battery type and other features might be engineered for simplicity and cost-effectiveness, but generally embodies the same concepts as in the preferred embodiment in which the second section is reused and/or recharged.

(8) In a preferred embodiment, the electronic smoking article **60** is about the same size as a conventional smoking article. In some embodiments, the electronic smoking article **60** can be about 80 mm to about 110 mm long, preferably about 80 mm to about 100 mm long and about 7 mm to about 8 mm in diameter. For example, in a preferred embodiment, the electronic smoking article is about 84 mm long and has a diameter of about 7.8 mm.

(9) Preferably, at least one adhesive-backed label is applied to the outer tube **6**. The label completely circumscribes the electronic smoking article **60** and can be colored and/or textured to provide the look and/or feel of a traditional smoking article. The label can include holes therein which are sized and positioned so as to prevent blocking of the air inlets **44**.

(10) The first section **70** includes an outer tube (or casing) **6** extending in a longitudinal direction and an inner tube (or chimney) **62** coaxially positioned within the outer tube **6**. Preferably, a nose portion **61** of an upstream gasket (or seal) **15** is fitted into an upstream end portion **65** of the inner tube **62**, while at the same time, an outer perimeter **67** of the gasket **15** provides a liquid-tight seal with an interior surface of the outer casing **6**. The upstream gasket **15** also includes a central, longitudinal air passage **20**, which opens into an interior of the inner tube **62** that defines a central channel **21**. A transverse channel **33** (shown in FIG. 2) at a backside portion of the gasket **15** intersects and communicates with the central channel **20** of the gasket **15**. This channel **33** assures communication between the central channel **20** and a space **35** (see FIG. 2) defined between the gasket **15** and a cathode connector piece **37**. In the preferred embodiment, the piece **37** includes a threaded section for effecting the threaded connection **205**.

(11) The outer tube **6** and/or the inner tube **62** may be formed of any suitable material or combination of materials. Examples of suitable materials include metals, alloys, plastics or composite materials containing one or more of those materials, or thermoplastics that are suitable for food or pharmaceutical applications, for example polypropylene, polyetheretherketone (PEEK), ceramic, and polyethylene. Preferably, the material is light and non-brittle.

(12) In the preferred embodiment, as shown in FIGS. 1 and 2, the electronic smoking article **60** includes at least one air inlet **44** formed in the outer tube **6**, preferably adjacent to the threaded connection **205** to minimize the chance of a smoker's fingers occluding one of the inlets and to control the resistance to draw (RTD) during smoking. In the preferred embodiment, the air inlets **44**, **44'** are sized and configured such that the electronic smoking article **60** has a RTD in the range of from about 60 mm H.sub.2O to about 150 mm H.sub.2O, more preferably about 90 mm H.sub.2O to about 110 mm H.sub.2O, most preferably about 100 mm H.sub.2O to about 130 mm H.sub.2O.

(13) In the preferred embodiment, the second section **72**, includes an air inlet **45** at an upstream end **5** of the electronic smoking article **60**, which is sized just sufficient to assure proper operation of the puff sensor **16**, located nearby. Drawing action upon the mouth end insert **8** is communicated to the air inlet port **45** through central channels provided in the anode post **47c** of the first section **70** and the anode connection post **47b** of the second section **72** and along space **13** between the battery **1** and the casing of the second section **72**. The air inlet port **45** is sized such that the airflow rate therethrough is much smaller than the airflow rates through the air inlets **44**, **44'**, so that the impact

on RTD is minimized and consistency in RTD is maintained. For example, each air inlet **44**, **44'** can be less than about 2.0 mm in width and less than about 1.5 mm in length.

(14) Preferably, a nose portion **93** of a downstream gasket **10** is fitted into a downstream end portion **81** of the inner tube **62**. An outer perimeter **82** of the gasket **10** provides a substantially liquid-tight seal with an interior surface **97** of the outer casing **6**. The downstream gasket **10** includes a central channel disposed between the central passage **21** of the inner tube **62** and the interior of the mouth end insert **8** and which communicates aerosol from the central passage **21** to the mouth end insert **8**.

(15) In the preferred embodiment, the liquid supply region **22** is contained in an outer annulus between inner tube **62** and outer tube **6** and between the gaskets **10** and **15**. Thus, the liquid supply region **22** at least partially surrounds the central air passage **21**. The liquid supply region **22** comprises a liquid material and optionally a liquid storage medium operable to store the liquid material therein.

(16) Preferably, the liquid storage medium is a fibrous material comprising cotton, polyethylene, cellulose, cellulose acetate, polyester, rayon and combinations thereof. Preferably, the fibers have a diameter ranging in size from about 6 microns to about 15 microns (e.g., about 8 microns to about 12 microns or about 9 microns to about 11 microns). The liquid storage medium can be a fibrous, sintered, porous, sponge, or foamed material. Also preferably, the fibers are sized to be irrespirable and can have a cross-section which has a round and/or hollow, y shape, cross shape, clover shape or any other suitable shape. In one embodiment, the liquid storage medium may comprise a winding of cotton gauze or other fibrous material about the inner tube **62**. In the alternative, the liquid supply region **22** may comprise a filled tank lacking a fibrous storage medium and containing only liquid material.

(17) Also preferably, the liquid material has a boiling point suitable for use in the electronic smoking article **60**. If the boiling point is too high, the heater-wick element **14** will not be able to vaporize the liquid. However, if the boiling point is too low, the liquid may vaporize even when the heater-wick element **14** is not activated.

(18) Preferably, the liquid material includes a tobacco-containing material including volatile tobacco flavor compounds which are released from the liquid upon heating. The liquid may also be a tobacco flavor containing material or a nicotine-containing material. Alternatively, or in addition, the liquid may include a non-tobacco material and/or a nicotine-free material. For example, the liquid may include water, solvents, ethanol, plant extracts and natural or artificial flavors. Preferably, the liquid further includes an aerosol former. Examples of suitable aerosol formers are glycerine and propylene glycol.

(19) In use, liquid material is transferred from the liquid supply region **22** and/or liquid storage medium via the heater-wick element **14**, which includes at least one wicking portion **140** and a heatable portion **141**. In the preferred embodiment, the heater-wick element **14** includes two wicking portions **140** and a heatable portion **141** therebetween. Also preferably, the wicking portions **140** and the heatable portion **141** are integrally formed of the same material. Thus, the heater-wick element **14** is a single piece of material.

(20) Advantageously, the heater-wick element **14** serves as both a wicking medium and a heating element. Thus, the heater-wick element **14** is a single piece structure and there is no need for a separate wick and heater. As such, the heater-wick element **14** is easy to manufacture and inexpensive as compared to electronic smoking articles including a separate heater and wick. Moreover, electricity flowing through the heater-wick element **14** can heat the heater-wick element via electrical resistance thereby heating the liquid material and lowering the viscosity of the liquid material, which can improve the flow of the liquid material from the wicking portions to the heatable portions of the heater-wick element.

(21) As shown in FIGS. **2** and **3**, the heater-wick element **14** includes a plurality of small metal beads or particles that have been fused (sintered) together. In an alternative embodiment, the beads

or particles can be glued together with a ceramic paste or other temperature resistant and potentially electrically conductive substance. The glue need not be electrically conductive. Each bead or particle is of a sub-millimeter diameter. Because a plurality of small metal beads or particles are fused together to form the heater-wick element **14**, the heater-wick element **14** includes internal cavities through which liquid material can travel by capillary action.

(22) As used herein, the term “particle” refers to beads, bits, rods, granules, powder, and pieces of any shape that can be fused together to form the heater-wick element **14** described herein.

(23) The heater-wick element **14** can be formed as a rod, a spiral, a block, a cylinder or a ribbon of metal beads or particles. Preferably, the heater-wick element **14** is substantially rigid. Moreover, the rod or ribbon can be straight, curved, or otherwise shaped to fit within the electronic cigarette.

(24) For example, the heater-wick element **14** can be U-shaped such that the heatable portion **141** is substantially straight and the wicking portions **140** extend upwardly or downwardly into the liquid supply region **22**, as shown in FIG. **3**. To position irregularly shaped heater-wick elements **14**, a slit can be made in each side of the inner tube **62** and extending to each slot. The heater-wick element **14** can be slid into place such that the heater-wick element **14** extends across the central air passage **21** and into the liquid supply region **22**.

(25) In another embodiment, as shown in FIG. **5**, the heater-wick element **14** can be H-shaped. The H-shaped heater-wick element **14** can include four wicking portions **140** extending into the liquid supply region and a heatable portion **141** extending across the central channel **21** of the inner tube **62**. Advantageously, an H-shaped heater-wick element **14** facilitates capillary draw of the liquid due to the use of four wicking portions **140**.

(26) Preferably, the heater-wick element **14** is substantially uniform in diameter and/or width at least along the heatable portion **141** of the heater-wick element **14**. Such uniformity promotes even heating. Alternatively, the heater-wick element **14** can vary in diameter and/or width along the length thereof so as to alter the heating profile of the heater-wick element **14**.

(27) As shown, the heater-wick element **14** can extend across the central channel **21** between opposing portions of the liquid supply region **22** and into the liquid supply region **22**. Thus, the wicking portion **140** at each end of the heater-wick element **14** extends through slots in the inner tube **62** and into the liquid supply region **22** so as to wick liquid into the heatable portion **141** of the heater-wick element **14**, which is positioned within the central air passage **21**. A closure ring can slide over an outer surface of the inner tube so as to substantially close off a remainder of open space provided between the heater-wick element and the slot, as described in U.S. Patent Application Publication No. 2013/0192619 filed Jan. 14, 2013, the entire content of which is incorporated herein by reference thereto.

(28) Preferably, the heater-wick element **14** is formed of a plurality of small metal beads or particles. Also preferably, the metal is an electrically conductive metal and the heater-wick element **14** is capable of withstanding repeated heating up to at least about 600° C. The size of the metal beads, the packing density of the metal beads and the type of metal are chosen to attain a targeted electrical resistance with high chemical resistance, good heating-induced degradation resistance and a low cost per heater-wick element **14**. Moreover, the bead size, density, and porosity can be varied along the length of the heater-wick element to attain a desired wicking and/or heating profile.

(29) Examples of suitable electrically resistive materials include titanium, zirconium, tantalum and metals from the platinum group. Examples of suitable metal alloys include stainless steel, nickel-, cobalt-, chromium-, aluminium- titanium- zirconium-, hafnium-, niobium-, molybdenum-, tantalum-, tungsten-, tin-, gallium-, manganese- and iron-containing alloys, and super-alloys based on nickel, iron, cobalt, stainless steel. For example, the heater-wick element **14** can be formed of nickel aluminides, a material with a layer of alumina on the surface, iron aluminides and other composite materials, the electrically resistive material may optionally be embedded in, encapsulated or coated with an insulating material or vice-versa, depending on the kinetics of

energy transfer and the external physicochemical properties required.

(30) Preferably, the heater-wick element **14** comprises at least one material selected from the group consisting of stainless steel, copper, copper alloys, nickel-chromium alloys, superalloys and combinations thereof. In a preferred embodiment, the heater-wick element **14** is formed of nickel-chromium alloys or iron-chromium alloys.

(31) In another embodiment, the heater-wick element **14** may be constructed of an iron-aluminide (e.g., FeAl or Fe.sub.3Al), such as those described in commonly owned U.S. Pat. No. 5,595,706 to Sikka et al., or nickel aluminides (e.g., Ni.sub.3Al). Use of iron-aluminides is particularly advantageous in that they exhibit high resistivity. FeAl exhibits a resistivity of approximately 180 micro-ohms, whereas stainless steel exhibits approximately 50 to 91 micro-ohms. The higher resistivity lowers current draw or load on the power supply (battery) **1**. In other embodiments, the heater-wick element **14** could comprise a metal particles or beads and ceramic particles or beads. In still other embodiments, the heater-wick element **14** is ceramic-free.

(32) Preferably, as shown in FIGS. **2**, **3**, and **4**, a brazed connection region (e.g., a post) **99** formed of a low-resistance material is brazed to each end or at two locations along a portion of the of the heater-wick element **14**. Preferably, the brazed connection regions **99** are formed just inside of the inner tube **62** and the heatable portion **141** extends between the brazed connection regions **99**. In another embodiment, the brazed connection regions **99** are contained entirely in the outer annulus as shown in FIG. **3**. By forming the brazed connection regions **99**, the electrical current is uniform across the length and width of heatable portion **141** of the heater-wick element **14** so as to avoid hot spots.

(33) For example, the brazed connection region **99** can be formed by wrapping a gold-plated wire around the heater-wick element **14** at select locations and brazing the wire to the heater-wick element **14** at selected locations so as to form a heatable portion **141** between the brazed connection regions **99**. Electrical leads **26** are attached to each brazed connection region **99** (or post), as shown in FIG. **4**, such that, when voltage is applied by the power supply, the heatable portion **141** heats the liquid material in the heatable portion **141** to a temperature sufficient to at least partially volatilize the liquid and form an aerosol. Alternatively, the electrical leads **26** can be attached directly to the heater-wick element **14** by sintering the electrical lead **26** directly into the heater-wick element **14**.

(34) In the preferred embodiment, the heater-wick element **14** is formed of a thermally and/or electrically conductive material. Suitable materials for forming the heater-wick element **14** are selected from the group consisting of stainless steel, copper, copper alloys, Inconel® available from Special Metals Corporation, which is a nickel-chromium alloy, Nichrome®, which is also a nickel-chromium alloy, and combinations thereof.

(35) In a preferred embodiment, the heater-wick element **14** is constructed from a plurality of small metal beads and/or particles each having a diameter of less than about 1 mm, less than about 0.5 mm or less than about 0.25 mm. Preferably, each of the beads or particles is substantially uniform in size. In other embodiment, the beads or particles can vary in size.

(36) The heater-wick element **14** has a length in the range of about 10 mm to about 15 mm, preferably about 12 mm or less, and a width in the range of about 0.5 mm to about 2.0 mm, preferably about 1.5 mm or less. Preferably, the heater-wick element **14** is placed in a transverse direction within the electronic smoking article. In other embodiments, other orientations are possible.

(37) Also preferably, the heater-wick element **14** achieves an electrical resistance ranging from about 0.3 Ohm to about 10 Ohms, more preferably about 0.8 Ohm to about 5.0 Ohms, more preferably about 4.0 Ohms or less.

(38) In addition, liquid can be drawn into the interstices, pores and/or voids between the metal beads and/or particles that form the heater-wick element **14**. Thus, the liquid moves along the heater-wick element from the wicking portions **140** to the heatable portion **141**. Moreover, the

heater-wick element **14** has a porosity of from about 20% to about 80%, more preferably about 30% to about 60% or about 40% to about 50%.

(39) Advantageously, the liquid material in the liquid supply region **22** is protected from oxygen (because oxygen cannot generally enter the liquid supply region **22** via the heater-wick element **14**). In some embodiments, the liquid material is also protected from light so that the risk of degradation of the liquid material is significantly reduced. Thus, a high level of shelf-life and cleanliness can be maintained.

(40) In the preferred embodiment, the liquid supply region **22** is sized and configured to hold enough liquid material such that the electronic smoking article **60** is operable for smoking for at least about 200 seconds, preferably at least about 250 seconds, more preferably at least 300 seconds and most preferably at least about 350 seconds. Thus, liquid supply region **22** is equivalent to about one pack of traditional smoking articles. Moreover, the electronic smoking article **60** can be configured to allow each puff to last a maximum of about 5 seconds.

(41) As shown in FIG. 2, the first section **70** can include a mouth end insert **8** having at least two diverging outlets **24** (e.g., 3, 4, 5 or more, preferably 2 to 10 outlets or more, more preferably 6 to 8 outlets, even more preferably 2 to 6 outlets or 4 outlets). Preferably, the outlets **24** are located off-axis and are angled outwardly in relation to the central channel **21** of the inner tube **62** (i.e., divergently). Also preferably, the mouth end insert (or flow guide) **8** includes outlets **24** uniformly distributed about the perimeter of mouth end insert **8** so as to substantially uniformly distribute aerosol in a smoker's mouth during use and create a greater perception of fullness in the mouth. Thus, as the aerosol passes into a smoker's mouth, the aerosol enters the mouth and moves in different directions so as to provide a full mouth feel. In contrast, electronic smoking articles having a single, on-axis orifice tend to direct its aerosol as single jet of greater velocity toward a more limited location within a smoker's mouth.

(42) In addition, the diverging outlets **24** are arranged and include interior surfaces **83** such that droplets of unaerosolized liquid material, if any, that may be entrained in the aerosol impact the interior surfaces **83** of the mouth end insert **8** and/or impact portions of walls **305** which define the diverging outlets **24**. As a result such droplets are substantially removed or broken apart, to the enhancement of the aerosol.

(43) In the preferred embodiment, the diverging outlets **24** are angled at about 5° to about 60° with respect to the longitudinal axis of the outer tube **6** so as to more completely distribute aerosol throughout a mouth of a smoker during use and to remove droplets. In a preferred embodiment, there are four diverging outlets **24** each at an angle of about 40° to about with respect to the longitudinal axis of the outer tube **6**, more preferably about 40° to about and most preferably about 42°.

(44) Preferably, each of the diverging outlets **24** has a diameter ranging from about 0.015 inch to about 0.090 inch (e.g., about 0.020 inch to about 0.040 inch or about 0.028 inch to about 0.038 inch). The size of the diverging outlets **24** and the number of diverging outlets **24** can be selected to adjust the resistance to draw (RTD) of the electronic smoking article **60**, if desired.

(45) The mouth end insert **8** may be integrally affixed within the tube **6** of the cartridge **70**. Moreover, the mouth end insert **8** can be formed of a polymer selected from the group consisting of low density polyethylene, high density polyethylene, polypropylene, polyvinylchloride, polyetheretherketone (PEEK) and combinations thereof. The mouth end insert **8** may also be colored if desired.

(46) In the preferred embodiment, the power supply **1** includes a battery arranged in the electronic smoking article **60** such that the anode **47a** is downstream of the cathode **49a**. A battery anode post **47b** of the second section **72** preferably contacts the battery anode **47a**.

(47) More specifically, electrical connection between the anode **47a** of the battery **1** and the heater-wick element **14** in the first section **70** is established through a battery anode connection post **47b** in the second section **72** of the electronic smoking article **60**, an anode post **47c** of the cartridge **70**

and an electrical lead **47d** connecting a rim portion of the anode post **47c** with the heater-wick element **14**. Likewise, electrical connection between the cathode **49a** of the battery **1** and the other lead of the heater-wick element **14** is established through the threaded connection **205** between a cathode connection fixture **49b** of the second section **72** and the cathode connector piece **37** of the first section **70** and from there through an electrical lead **49c** which electrically connects the fixture **37** to the opposite lead of the heater-wick element **14**.

(48) The battery can be a Lithium-ion battery or one of its variants, for example a Lithium-ion polymer battery. Alternatively, the battery may be a Nickel-metal hydride battery, a Nickel cadmium battery, a Lithium-manganese battery, a Lithium-cobalt battery or a fuel cell. In that case, preferably, the electronic smoking article **60** is usable by a smoker until the energy in the power supply is depleted or in the case of lithium polymer battery, a minimum voltage cut-off level is achieved.

(49) Alternatively, the power supply **1** may be rechargeable and include circuitry allowing the battery to be chargeable by an external charging device. In that case, preferably the circuitry, when charged, provides power for a pre-determined number of puffs, after which the circuitry must be re-connected to an external charging device. To recharge the electronic smoking article **60**, a USB charger or other suitable charger assembly can be used.

(50) Preferably, the electronic smoking article **60** also includes control circuitry including a puff sensor **16**. The puff sensor **16** is operable to sense an air pressure drop and initiate application of voltage from the power supply **1** to the heater-wick element **14**. As shown in FIG. 2, the control circuitry can also include a heater activation light **48** operable to glow when the heatable portion **141** of the heater-wick element **14** is activated. Preferably, the heater activation light **48** comprises an LED and is at an upstream end of the electronic smoking article **60** so that the heater activation light **48** takes on the appearance of a burning coal during a puff. Moreover, the heater activation light **48** can be arranged to be visible to the smoker. In addition, the heater activation light **48** can be utilized for smoking article system diagnostics or to indicate that recharging is in progress. The light **48** can also be configured such that the smoker can activate and/or deactivate the light **48** for privacy, such that the light **48** would not activate during smoking if desired.

(51) Preferably, the at least one air inlet **45** (FIG. 1) is located adjacent the puff sensor **16**, such that the puff sensor **16** senses air flow indicative of a smoker taking a puff and activates the power supply **1** and the heater activation light **48** to indicate that the heatable portion **141** of the heater-wick element **14** is working.

(52) A control circuit is preferably integrated with the puff sensor **16** and supplies power to the heater-wick element **14** responsive to the puff sensor **16**, preferably with a maximum, time-period limiter.

(53) Alternatively, the control circuitry may include a manually operable switch for a smoker to initiate a puff. The time-period of the electric current supply to the heater-wick element may be pre-set depending on the amount of liquid desired to be vaporized. Alternatively, the circuitry may supply power to the heater-wick element **14** as long as the puff sensor **16** detects a pressure drop.

(54) Preferably, when activated, the heater-wick element **14** heats and volatilizes liquid in contact with the heater-wick element **14** for less than about 10 seconds, more preferably less than about 7 seconds. Thus, the power cycle (or maximum puff length) can range in period from about 2 seconds to about 10 seconds (e.g., about 3 seconds to about 9 seconds, about 4 seconds to about 8 seconds or about 5 seconds to about 7 seconds).

(55) When the word “about” is used in this specification in connection with a numerical value, it is intended that the associated numerical value include a tolerance of $\pm 10\%$ around the stated numerical value. Moreover, when reference is made to percentages in this specification, it is intended that those percentages are based on weight, i.e., weight percentages.

(56) Moreover, when the words “generally” and “substantially” are used in connection with geometric shapes, it is intended that precision of the geometric shape is not required but that

latitude for the shape is within the scope of the disclosure. When used with geometric terms, the words “generally” and “substantially” are intended to encompass not only features which meet the strict definitions but also features which fairly approximate the strict definitions.

(57) It will now be apparent that a new, improved, and nonobvious electronic smoking article has been described in this specification with sufficient particularity as to be understood by one of ordinary skill in the art. Moreover, it will be apparent to those skilled in the art that numerous modifications, variations, substitutions, and equivalents exist for features of the electronic smoking article which do not materially depart from the spirit and scope of the invention. Accordingly, it is expressly intended that all such modifications, variations, substitutions, and equivalents which fall within the spirit and scope of the invention as defined by the appended claims shall be embraced by the appended claims.

Claims

1. An electronic vaping cartridge comprising: a liquid supply region including liquid material; an air channel extending through the liquid supply region; a heater-wick element including a heatable portion and a wicking portion, the heater-wick element formed of a plurality of particles, the heater-wick element in fluidic communication with the liquid material and oriented in a transverse direction across the air channel; and a connection structure wrapped around the heater-wick element and configured to connect the heater-wick element to electrical leads.
 2. An electronic vaping cartridge comprising: a liquid supply region including liquid material; an air channel extending through the liquid supply region; and a heater-wick element including a heatable portion and a wicking portion, the heater-wick element formed of a plurality of particles, the heater-wick element in fluidic communication with the liquid material and oriented in a transverse direction across the air channel, wherein the wicking portion extends into the liquid supply region, and wherein the wicking portion includes a first wicking portion, a second wicking portion, a third wicking portion, and a fourth wicking portion, the first wicking portion and the second wicking portion disposed at one end of the heatable portion, and the third wicking portion and the fourth wicking portion disposed at an opposing end of the heatable portion.
 3. An electronic vaping cartridge comprising: a liquid supply region including liquid material; an air channel extending through the liquid supply region; and a heater-wick element including a heatable portion and a wicking portion, the heater-wick element formed of a plurality of particles, the heater-wick element in fluidic communication with the liquid material and oriented in a transverse direction across the air channel, wherein the plurality of particles are metal beads, and wherein the metal beads are glued together with a ceramic paste.
 4. The electronic vaping cartridge of claim 1, wherein the plurality of particles include a ceramic material.
 5. The electronic vaping cartridge of claim 1, wherein the plurality of particles are formed of at least one of a nickel aluminide or an iron aluminide.
 6. The electronic vaping cartridge of claim 1, wherein the plurality of particles are fused together.
 7. The electronic vaping cartridge of claim 1, wherein the heater-wick element includes an insulating material that coats the plurality of particles.
 8. The electronic vaping cartridge of claim 1, wherein the heater-wick element includes an insulating material that is embedded in the plurality of particles.
 9. The electronic vaping cartridge of claim 1, wherein the heater-wick element is U-shaped.
 10. The electronic vaping cartridge of claim 1, wherein the heater-wick element varies in cross-section along a length thereof.
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